

Kinetic Theory of Gas

Basic Assumptions

The kinetic theory of gases is based upon the following simplifying fundamental assumptions or postulates:

- i. The kinetic theory assumes that a gas consists of an exceedingly large number of minute particles all having the same mass and obeying Newton's laws of motion.
- ii. The molecules are supposed to be perfectly elastic spheres or mass points. The volume occupied by the molecules may be neglected in comparison with the volume of the container of the gas. These molecules can experience elastic collision with one another and with the walls of the container.
- iii. The molecules are continuously in motion with varying velocities and the molecules move in straight lines between any two consecutive collisions.
- iv. The molecules of the gas are sufficiently far apart most of the time so that they do not exert any force of attraction or repulsion between them. Therefore, they cannot have any potential energy, but have only kinetic energy.
- v. The dimension/size of a molecule may be neglected in comparison with the distance traversed by it between two successive collisions, called its free path.
- vi. The time of impact is negligible in comparison to the time taken to traverse the free path.
- vii. The distribution of velocities among the gas molecules at equilibrium does not change with time. The same number of molecules always has a particular velocity, although the specific molecules having this velocity may be different at different times.

Pressure Exerted by a Perfect Gas

The pressure of the gas is due to the impacts of the molecules upon the walls of the container and is equal to the rate of change of momentum over unit area.

$$P = \frac{F}{A} = \frac{mv}{tA}$$

Let us consider a cubic vessel ABCDEFGH of side l cm containing the gas. The volume of the vessel and hence that of the gas is l^3 cc. Let n and m represent the very large number of molecules present in the vessel and the mass of each molecule, respectively.

Let us consider a molecule P moving in a random direction with a velocity C_1 . The velocity can be resolved into three perpendicular components u_1 , v_1 and w_1 along the X, Y and Z axis, respectively.

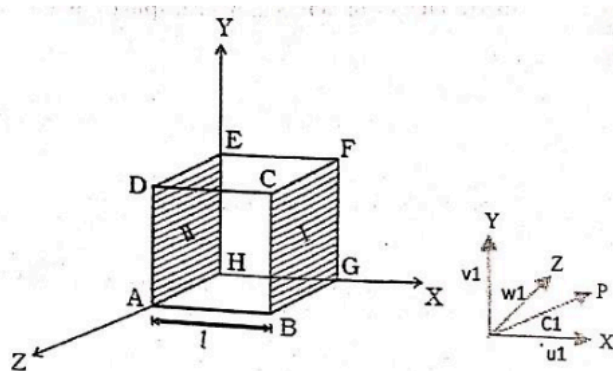


Fig. 3.1

Therefore,

$$C_1^2 = u_1^2 + v_1^2 + w_1^2$$

The component of the velocity with which the molecule P will strike the opposite face BCFG is u_1 and the momentum of the molecule is mu_1 . This molecule is reflected back with the same momentum mu_1 , in an opposite direction and after traversing a distance, l will strike the opposite face ADEH.

The change in momentum produced due to the impact is $mu_1 - (-mu_1) = 2mu_1$.

As the velocity of the molecule is u_1 , the time interval between two successive impacts on the wall BCFG is $\frac{2l}{u_1}$ seconds.

Change in momentum produced per second due to the impact of this molecule is

$$\frac{2mu_1}{2l/u_1} = 2mu_1 \times \frac{u_1}{2l} = \frac{mu_1^2}{l}$$

The force F_x , due to the impact of all the n molecules in one second $= \frac{m}{l} [u_1^2 + u_2^2 + \dots + u_n^2]$

Force per unit area on the wall BCFG or ADEH is equal to the pressure P_x

$$P_x = \frac{m}{l \times l^2} [u_1^2 + u_2^2 + \dots + u_n^2] = \frac{m}{l^3} [u_1^2 + u_2^2 + \dots + u_n^2]$$

Similarly, the pressure P_y on the walls CDEF or ABGH is given by

$$P_y = \frac{m}{l^3} [v_1^2 + v_2^2 + \dots + v_n^2]$$

And the pressure P_z on the walls ABCD or EFGH is given by

$$P_z = \frac{m}{l^3} [w_1^2 + w_2^2 + \dots + w_n^2]$$

As the pressure of a gas is the same in all directions, the mean pressure P is given by

$$\begin{aligned} P &= \frac{P_x + P_y + P_z}{3} \\ &= \frac{m}{3l^3} [(u_1^2 + v_1^2 + w_1^2) + (u_2^2 + v_2^2 + w_2^2) + \dots + (u_n^2 + v_n^2 + w_n^2)] \\ &= \frac{m}{3l^3} [C_1^2 + C_2^2 + \dots + C_n^2] \dots \dots \dots (3.1) \end{aligned}$$

But volume, $V = l^3$. Let C be the root mean square (R.M.S.) velocity of the molecules.

Then

$$\begin{aligned} C^2 &= \frac{C_1^2 + C_2^2 + \dots + C_n^2}{n} \\ \therefore nC^2 &= C_1^2 + C_2^2 + \dots + C_n^2 \end{aligned}$$

Substituting this value in equation (3.1), we get,

$$P = \frac{mnC^2}{3V} \dots \dots \dots (3.2)$$

But $M = mn$, where M is the mass of the gas of volume V , m is the mass of each molecule and n is the number of molecules in a volume V .

$$\begin{aligned} P &= \frac{MC^2}{3V} \\ P &= \frac{1}{3} \rho C^2 \dots \dots \dots (3.3) \end{aligned}$$

where, ρ the density of the gas.

The equation (3.3) is known as Bernoulli's equation.

Again

$$P = \frac{1}{3} \rho C^2 = \frac{2}{3} \times \frac{1}{2} \frac{M}{V} C^2 = \frac{2}{3} E \dots \dots \dots (3.4)$$

where E is the mean kinetic energy of translation of the molecules per unit volume.

The pressure is, therefore two-thirds of the total kinetic energy of translation of the molecules in unit volume.

From equation (3.3),

$$C^2 = \frac{3P}{\rho}$$

$$C = \sqrt{\frac{3P}{\rho}} \text{ [C is the R. M. S. velocity]}$$

Kinetic Interpretation of Temperature

The pressure exerted by a gas is

$$P = \frac{1}{3} \rho C^2 = \frac{1}{3} \frac{M}{V} C^2$$

$$\therefore PV = \frac{1}{3} MC^2$$

Let us consider 1 gm molecule of the gas at a temperature T K.

$$PV = RT$$

$$\frac{1}{3} MC^2 = RT$$

$$\frac{1}{2} MC^2 = \frac{3}{2} RT \dots \dots \dots (3.5)$$

This equation gives a physical meaning to the gas constant R and shows that it is equal to two-thirds of the total translational energy of the molecules in one gm molecule of the gas.

Let the mass of each molecule be m and Avogadro's number be N.

Then,

$$M = mN$$

$$\frac{1}{2} mNC^2 = \frac{3}{2} RT$$

$$\text{or, } \frac{1}{2} mC^2 = \frac{3R}{2N} T = \frac{3}{2} kT \dots \dots \dots (3.6)$$

Here k is the Boltzmann's constant.

LOOK

Equation of State of an Ideal Gas

From kinetic theory,

$$P = \frac{1}{3} \rho C^2 = \frac{1}{3} \frac{M}{V} C^2$$
$$\therefore PV = \frac{1}{3} MC^2$$

Consider one gram molecule of a gas at an absolute temperature T.

The mean energy of the molecules $= \frac{1}{2} MC^2 = \frac{1}{2} mNC^2$

$$\therefore PV = \frac{1}{3} mNC^2$$
$$= \frac{2}{3} N \frac{1}{2} mC^2$$

Mean kinetic energy of a molecule $= \frac{1}{2} mC^2 = \frac{3}{2} kT$

Hence,

$$PV = \frac{2}{3} N \frac{3}{2} kT = NkT$$

But, $N \times k = R$

$$\therefore PV = RT$$

where P is in dynes/sq. cm., $R = 8.31 \times 10^7$ erg/g mole.K, T is in K, V is the volume in cc per gram molecule.

Brownian Motion

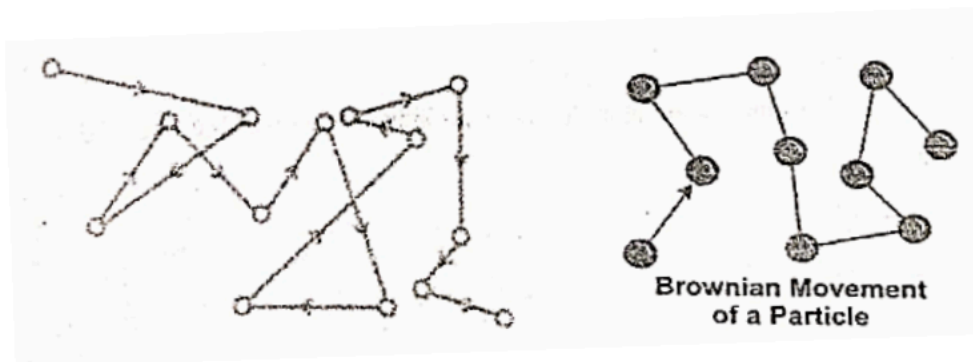


Fig. 3.2: Brownian motion.

If small particles suspended in a liquid, as in a colloidal solution, and illuminated by a strong light be examined under a powerful microscope, the particles appear to move to and fro rapidly and continuously in an irregular manner. Each particle spins, rises, sinks and rises again. This haphazard motion is known as Brownian movement or Brownian motion.

The following important facts about the Brownian motion have been subsequently observed by various scientists as a result of experimental study:

- i. The motions of the particles are continuous. This movement is attributed to the impact of the molecules of the liquids on the suspended particles. Particles, because they are small in size, are acted on by an unbalanced force due to impact and they are in motion. Thus, Brownian motion is a direct proof of the existence of molecular motion.
- ii. The smaller the particles, the quicker are the motions.
- iii. They are independent of the shaking of the vessel.
- iv. The lower the viscosity of the liquid, the quicker and the more chaotic are the motions.
- v. Two particles of the same size move equally fast at the same temperature.

The phenomenon of Brownian movement gives a clear picture of the gaseous state of matter. The motion of the molecules of a gas is similar in nature to the Brownian motion of suspended particles in colloidal solution.

Degrees of Freedom

The number of directions in which a particle can move freely is defined as its degrees of freedom. Or, the number of independent coordinates required to specify completely the position and orientation of a particle in space, is defined as its degrees of freedom (DOF). A molecule in a gas can move along any of the three co-ordinate axes. It has three degrees of freedom. The DOF is a way in which an atom or molecule can store energy.

Monoatomic molecule: A monoatomic gas molecule (like He, Ne etc.) has three translational DOF and zero rotational DOF. Hence its DOF is $3 + 0 = 3$.

In a more general way, we shall define number of degrees of freedom of a dynamical system as the number of independent squared terms which enter into the expression for the energy of the system. The kinetic energy of a particle which is capable of translation but not of rotation, such as a molecule of a monoatomic gas, is given by $\frac{1}{2}mu^2 + \frac{1}{2}mv^2 + \frac{1}{2}mw^2$.

Since there are three independent squared terms in the expression of energy, so each particle has three degrees of freedom.

Diatomic molecule: At low temperature (below 100K), a diatomic gas molecule (H_2 , O_2 , N_2 etc.) has 3 DOF due to its translational motion. At moderate temperatures ($100K < T < 1000K$) diatomic molecules have 5 DOF (3 translational + 2 rotational). At high temperatures (above 1000K) it has 7 DOF (3 translational + 2 rotational + 2 vibrational).

Triatomic molecule: A triatomic molecule, if it is non-linear (H_2O , H_2S , SO_2 etc.) has 6 DOF (3 translational + 3 rotational) and if it is linear (CO_2 , CH_2 etc.) has 5 DOF (3 translational + 2 rotational). These molecules also have vibrational DOF particularly at high temperature.

Principle of Equipartition of Energy

Statement: It states that for any dynamical system in thermal equilibrium, the total energy is divided equally among all the degrees of freedom and the energy associated per molecule per degree of freedom is $\frac{1}{2}kT$.

According to kinetic theory of gases, the mean kinetic energy of a molecule at a temperature T is given by

$$\frac{1}{2}mC^2 = \frac{3}{2}kT$$

$$\text{But } C^2 = u^2 + v^2 + w^2$$

As x, y and z are all equivalent, mean square velocities along the three axes are equal.

$$\therefore u^2 = v^2 = w^2$$

$$\text{or, } \frac{1}{2}mu^2 = \frac{1}{2}mv^2 = \frac{1}{2}mw^2$$

$$\therefore \frac{1}{2}mC^2 = 3 \left[\frac{1}{2}mu^2 \right] = 3 \left[\frac{1}{2}mv^2 \right] = 3 \left[\frac{1}{2}mw^2 \right] = \frac{3}{2}kT$$

$$\therefore \frac{1}{2}mu^2 = \frac{1}{2}kT \dots \dots \dots (3.7)$$

$$\frac{1}{2}mv^2 = \frac{1}{2}kT \dots \dots \dots (3.8)$$

$$\frac{1}{2}mw^2 = \frac{1}{2}kT \dots \dots \dots (3.9)$$

Therefore, the average kinetic energy associated with each degree of freedom = $\frac{1}{2}kT$

Thus, the energy associated with each degree of freedom (whether translatory or rotatory) is $\frac{1}{2}kT$. For a vibrational degree of freedom, since the energy is half kinetic and half potential, the total energy is $\frac{1}{2}kT + \frac{1}{2}kT = kT$, if we consider each of the vibrational modes as a single degree.